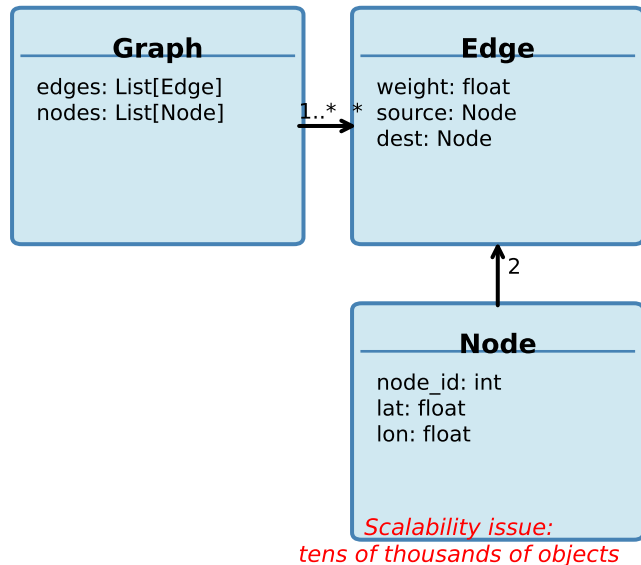


# Graph Data Structure Representations

Object-Oriented Design



Adjacency Matrix (compact, fast)

|    |    |    |    |    |
|----|----|----|----|----|
|    | j1 | j2 | j3 | j4 |
| j1 | 0  | 1  | 1  | 0  |
| j2 | 1  | 0  | 0  | 1  |
| j3 | 0  | 0  | 0  | 1  |
| j4 | 0  | 1  | 1  | 0  |
|    | j1 | j2 | j3 | j4 |

From ↓

To →

*O(1) lookup:  $A[i][j]$  tells us directly  
if j1 connects to j2*